Md Akib Haider

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Research Interests: Human-Centric AI, Computer Vision, Machine Learning

EDUCATION

- **Islamic University of Technology (IUT)** Aug 2022 – Oct 2026
  B.Sc. in Computer Science and Engineering
  - **CGPA**: 3.91/4.00

RESEARCH EXPERIENCE

- **Prototype-Based Test-Time Prediction Calibration for Few-Shot Object Detection**
  Accepted in WACV | Paper 2027
  - We proposed a Patch-Token Prototype Calibration method by encoding query images via a frozen foundation-model encoder for fully inference-based confidence recalibration of any CNN-based detector.
- **Diagnosing Prototype Estimation in Frozen-Feature Few-Shot Class-Incremental Learning: Training-Free Geometry Selection and Unlabeled-Query Prototype Rectification**
  Accepted in WACV | Paper 2027
  - We proposed finite-shot prototype estimation as the dominant recoverable error source in frozen few shot class-incremental learning, and complements it with a transductive correction for unlabeled query settings.
- **Probabilistic Feature Imputation and Uncertainty-Aware Multimodal Federated Aggregation**
  Published in PMLR | Paper 2026
  - We designed a probabilistic federated learning framework that replaces point estimates with calibrated distributions for uncertainty-aware local fusion and weighted aggregation across missing modalities.
- **LoRA-MME: Multi-Model Ensemble of LoRA-Tuned Encoders for Code Comment Classification**
  Accepted in ICSE workshop | Paper 2026
  - We proposed a weighted stacking ensemble learning method while fine-tuning four complementary code encoders to solve multi-label classification from code comments across different programming languages.
- **Semi-Supervised Frequency-Aware GAN for Ultra Low to High Field MRI Translation** | Under Review, ISBI
  - We presented a semi-supervised frequency-aware attention-guided framework for synthesizing 3T MRI from 64mT ultra-low-field scans, addressing paired data scarcity through adversarial refinement.

WORK EXPERIENCE

- **Undergraduate Research Intern** Jun 2026 – Present
  University of Illinois Urbana Champaign, SRSE program, Sharifa Sultana's Lab
- **App Design and Development (Remote)** Feb 2025 – Apr 2025
  SwadeshBazar, Chelmsford, UK
  - Built Android and web version in Flutter and Next.js for a grocery delivery and warehousing start-up featuring product browsing, cart, dashboard, designed with Rave and secure payment gateway.

PROJECTS

- **29 Game State Detection and Move Analysis using Video-Scene Understanding** 2026
  Python, PyTorch, OpenCV, OpenVINO, YOLOv26 | code
  - Built a system to detect game states, analyze gameplay, and predict moves in the card game 29 under low-resource settings.
- **Better_SIS, University Management System** | *Dart, C++, Swift, Flutter* | code 2025
  - Developed Student Information System (SIS) of IUT, featuring academic activities and internal service management for students, teachers, and staff.

- **Student Driven Market Place** | *Flutter, Gemini 2.5, Open Street Map* | code 2025
  - Created a virtual institutional marketplace where students can buy and sell their products within the community, donate items to those in need, and easily manage product deliveries.
- **Xchange, File Sharing System** | *Socket Programming, Java* | code 2024
  - Developed a secure and efficient peer-to-peer file-sharing system enabling fast file transfers over a local network without requiring internet access.

GRANTS AND FELLOWSHIPS

- **ACM SIGHPC Travel Grant**
  - HPC Immersion Program, Attended SC 2025 in St. Louis, Missouri, USA 2025
  - Lightning Talk on "Data, Model, and Tensor Parallelism for efficient MRI Translation".

ACHIEVEMENTS

Contests & Awards

- **NLBSE Tool Competition** winner, co-located workshop with ICSE-26. 2026
- **Programming Contest**
  - **26th** (among 110 teams), IUT 12th National ICT Fest Programming Contest. 2026
  - **6th**, Intra-IUT Junior Programming Contest. 2024
  - **52nd** (among 115 teams), IUT 11th National ICT Fest Programming Contest. 2024
  - Personal best: Country **rank 37**, Global **rank 1536** in Codeforces | profile. 2023-26
- **Hackathon**
  - **Champion** in Coderush 2.0, IUT-ICT Fest Hackathon | code 2025
- **Software Sub-Team Member, Project Altair** at IUT Robotics Society. 2023–24
  - **Global 6th place** and **Best Science Team award** in International Rover Challenge (IRC).

Games & Sports

- **Captain,** Inter University Chess Team of IUT | *FIDE Blitz Rating: 1553*. 2025–26
- **Champion,** Intra-IUT Annual Drought and Scrabble Competition. 2026
- **2nd Runner Up,** Intra-IUT Annual Table Tennis Competition. 2026
- All Bangladesh **rank 49** at 37th National Sub-Junior Chess Championship. 2018

LEADERSHIP & EXTRA CURRICULAR ACTIVITIES

- **Instructor, IUT ACM Community** Aug 2024 – Present
  - Mentor for sophomore Competitive Programming aspirants.
  - Organized and conducted weekly classes.
- **Organizer (Event Head), Intra-IUT Junior Programming Contest** 2025
- **Problem Setting**
  - Intra-IUT Junior Programming Contest | problemset 2025
  - Abid Hasan Intra University Programming Contest | problemset 2025

Sports: Passionate about playing table tennis and chess.

REFERENCES

- **Dr. Md. Hasanul Kabir**
  Professor, Department of CSE, Islamic University of Technology, [Thesis Supervisor]
  hasanul@iut-dhaka.edu
- **Dr. Sharifa Sultana**
  Assistant Professor, Department of CSE, University of Illinois Urbana-Champaign, [Research Supervisor]
  sharifas@illinois.edu
- **Dr. Rafsanjany Kushol**
  Postdoctoral Scholar, Faculty of Medicine, Medicine Department, University of Alberta [Research Supervisor]
  kushol@ualberta.ca